

Utilization of Electricity

Electrical Safety Practices & Risk Assessment

General Science | Quarter 1: Lesson 8.1 (Ref: SHS_GS_Q1_LAS_LE7.pdf)

The Power and the Peril

Electricity powers our modern world. But what happens when the massive energy we rely on escapes its intended, safe path?



Have you ever experienced a 'mild
shock' from a household appliance?
Why do we often ignore these critical
warning signs?

REFLECTION POINT - THINK, PAIR, SHARE



The Physics of the Hazard

Thermal Hazards

These occur when excessive electrical power causes rapid, undesired heating. This thermal energy transfer is the root cause of devastating electrical fires.

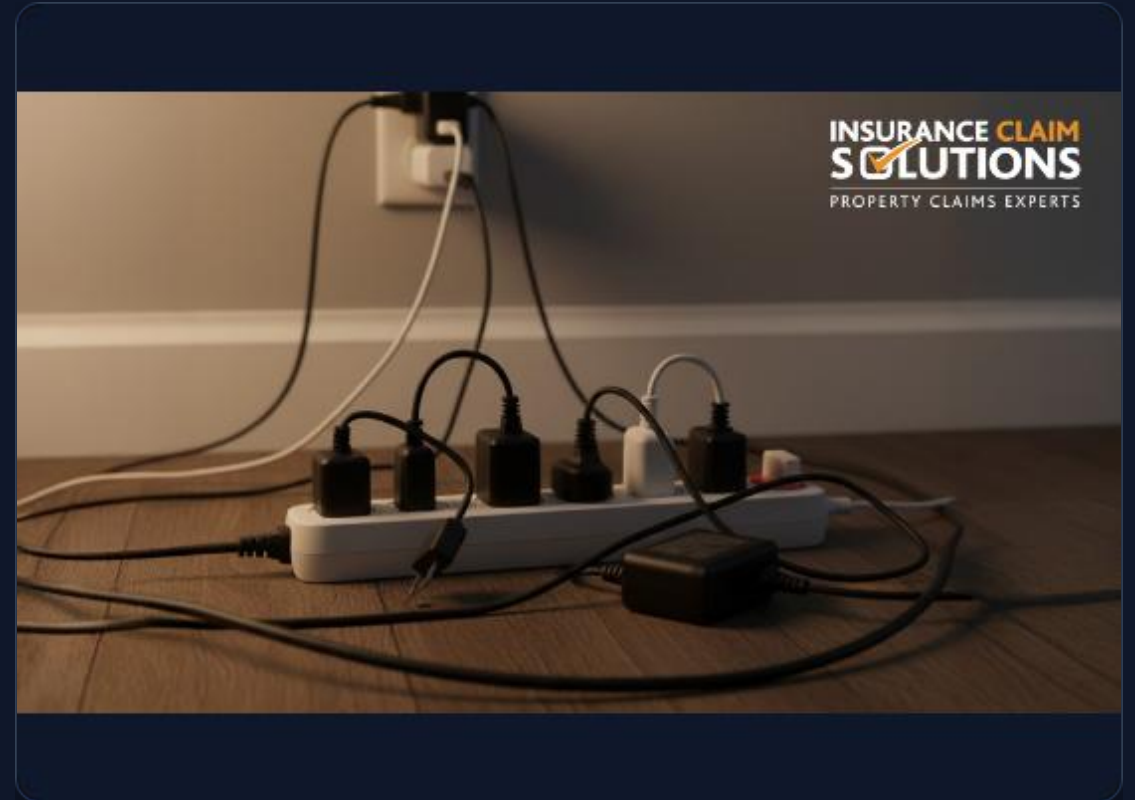
Shock Hazards

These occur when an electric current physically passes through a person. Shocks can range from a painful jolt to dangerous, heart-stopping lethality.

Thermal Hazards: Overloading

Plugging multiple appliances into one "octopus" outlet creates a dangerous parallel circuit.

This configuration draws massive amounts of current, overwhelming the wiring's safe capacity and generating intense heat.



The Math Behind the Fire

According to Joule's Law of Heating, thermal power (P) is related to Current (I) and Resistance (R):

$$P = I^2 R$$

Because thermal power increases with the **square** of the current, overloaded sockets generate heat incredibly fast, melting wire insulation.

Why is the **Body** a Target?

The Human Conductor

Humans are highly conductive due to water and electrolytes. If an appliance shorts, the electricity uses you as the path of least resistance to the ground.

The "Muscle Freeze"

Alternating current (AC) interferes with brain signals. At just 10–20mA, muscular contraction occurs. If you grab a live wire, you physically cannot let go.

Stop and Think!

In chemistry, we learn that pure, distilled water is actually a very *poor* conductor of electricity.
Given this fact, why are damp basements or operating an appliance with sweaty hands considered extreme, high-risk hazards?

Activity 1:

Hazard Hunt

It is time to test your knowledge.

Following your Activity Sheet
(SHS_GS_Q1_LAS_LE7.pdf), you will inspect
an area for potential hazards.

Document findings like a professional safety
inspector.



Phase 1: Spot the Danger



Outlets & Wires

Check for overloaded sockets, damaged wires, and flickering lights.



Appliances

Identify overheating devices, burning smells, or tangled cords.



Environment

Look for improper ventilation and electrical items near water.

Focus: Outlets & Wires

When inspecting your area, pay close attention to the integrity of the power delivery system:

- ⚡ Look for black scorch marks around wall plates.
- ⚡ Feel for unexpected warmth near outlets.
- ⚡ Note any lights that flicker when an appliance turns on.



Focus: Environmental Risks

Placement Matters

Wires stretched tightly across walkways are tripping hazards that can damage internal copper strands, leading to arcing and unseen fires.

Heat Traps

Large appliances pushed tightly against walls lack proper ventilation. This traps waste heat, degrading electrical insulation rapidly over time.

Phase 2: Power Up Safety

Risk Level	Description of Danger	Action Protocol
Low Risk	Unused devices left plugged in but turned off.	Manage easily. Unplug the device safely.
Medium Risk	Tangled cords or poor appliance ventilation.	Re-route wires to prevent future damage.
High Risk	Exposed live wires or overheating appliances.	Isolate the area & report to experts!

From Knowledge to **Action**



Qualified Repairs

Never fix high-risk hazards yourself. Call a professional.



Emergency Prep

Locate the nearest Class C fire extinguisher for electrical fires.



Daily Safe Habits

Always pull by the solid plug base, never yank the cord.

Synthesis & Reflection (Part 1)

 What were the most common electrical hazards you identified during your specific area inspection?

 Were there any hazards that completely surprised you or that you had simply never considered before?

Synthesis & Reflection (Part 2)



Based on your matrix, which hazards pose the absolute greatest risk to human life or property?



How did you arrive at your answers, and what actions can schools take to improve safety culture?

Questions?

Stay safe and stay curious.

Image Sources



<https://insuranceclaimsolutions.ie/wp-content/uploads/2025/09/a-photo-of-a-dangerously-overloaded-extension-socket.png>

Source: insuranceclaimsolutions.ie



https://cdn.prod.website-files.com/6244a527688c6005c7c25202/6830e04297567f6544cde18e_shutterstock_2456277121.jpg

Source: www.gotmold.ca



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